Activity 1.1.1 Over-Easy Engineering

OVER-EASY ENGINEERING

Accelerating Agricultural Advancements

GOALS | MATERIALS | RESOURCES

- Apply a design process to creatively solve a problem.
- Develop, test, and evaluate a potential solution to verify it meets constraints and criteria.
- Contribute to the efforts of a team.

GOALS | MATERIALS | RESOURCES

- Confetti eggs
- Popsicle sticks
- Rubber bands
- Cardboard
- Scissors

- Tape
- String
- Glue
- Egg basket

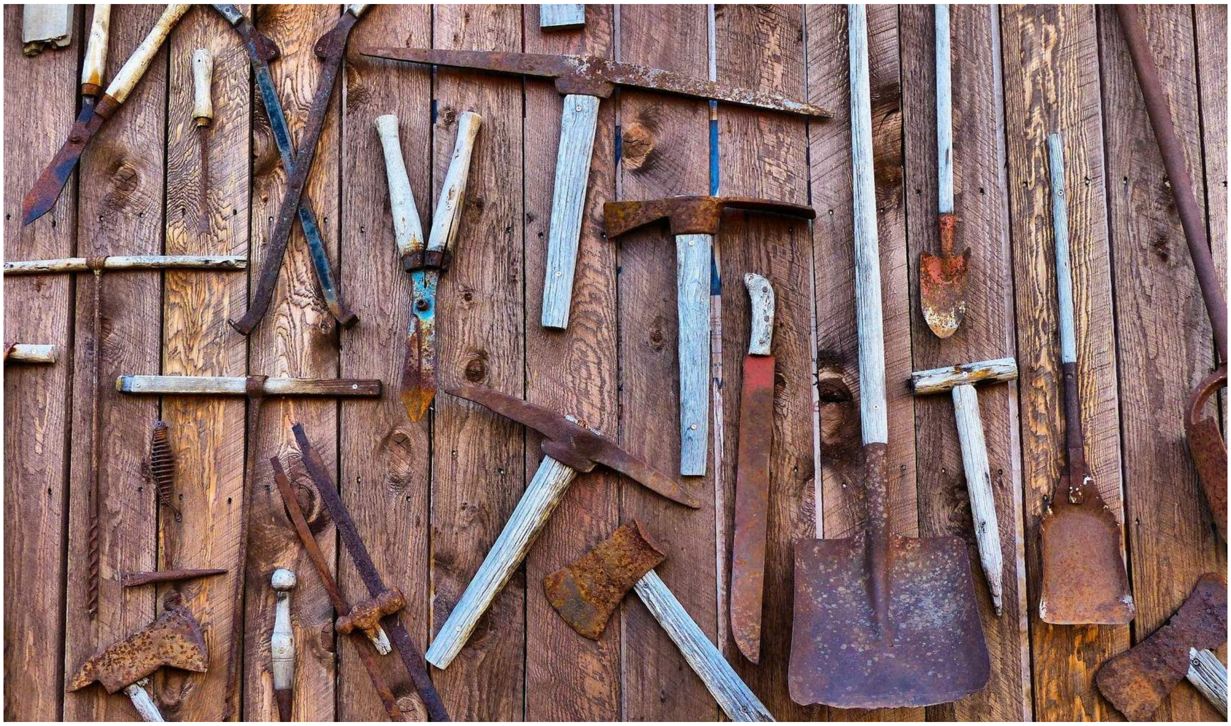
GOALS	MATERIALS	RESOURCES

- **Activity 1.1.1 Over-Easy Engineering (Downloadable PDF)**

Since the Stone Age, people have been creating and using tools to solve problems. Tools are used in simple daily tasks, like eating dinner with a fork and knife, as well as complex feats of engineering, like building the world's tallest structures with cranes. One environment that uses a variety of tools is agriculture.

Agriculture has been in a state of growth and innovation since people started farming with simple shovels and pitchforks. Over time, tractors and other mechanical devices were developed to help grow and collect crops that fill our grocery stores and local food stands.

In this challenge, you will create an expandable grabbing tool to help retrieve a small item from a hard-to-reach place on a farm.



Meet Fran the Farmer

Francesca is a mechanical engineering graduate who has decided to apply her engineering skills to help feed her country. Her goal is to increase efficiency, sustainability, and entrepreneurship in agriculture and farming. Francesca's efforts quickly earned her the nickname "Fran the Farmer."

Each morning, Fran helps collect eggs in chicken coops, and it is difficult to reach all of them. She wants to develop a tool to improve her morning routine so she can reach and collect eggs faster.

Reach for It!

1. With your team, review the challenge materials and instructions, make a plan, and get started!

Engineering Notebook

Document all stages of your work in your PLTW Engineering Notebook. Describe your ideas by making notes or sketching. List questions you have or problems you encounter.

INSTANT CHALLENGE

Your Goal: Using only the materials provided, design and build an expandable tool that can reach an egg a minimum of 10 inches away. Your tool should be able to open and close firmly around an egg. Your goal is to hold and move the largest and heaviest egg into your basket without breaking your tool or cracking the egg!

Unit 1 Problem Introduction

Fix Fran's Farm

There are seemingly endless tasks to complete each day on a farm, and a variety of tools are used for each task. Mechanical engineers constantly innovate to develop new tools to help farmers and solve agricultural problems, such as the following:

- How can we remove or prevent invasive insects from harming crops?
- Is there a more efficient way to plant a certain crop?
- What other ways can we use our tools to increase their versatility?

Consider these questions as you progress through Unit 1 to explore the world of agriculture and mechanical engineering. Each activity prepares you for solving the final unit problem of developing a mechanical device to solve an agricultural problem.

Conclusion

Question 1

How successful was your tool at solving this challenge? Support your answer with evidence you gathered during the challenge.

Question 2

How has your teamwork and collaboration improved since your first instant challenge? How can you continue to improve these skills?

Question 3

How has your application of the engineering design process improved since your first instant challenge? Where do you need the most improvement? Why?